DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT



Calculus and Optimization

Derivatives and Their Types

Lecture No. -02



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Recap of previous lecture



Topic

DERIVATIVES & THEIR TYPES

(PART-1)

Topics to be Covered



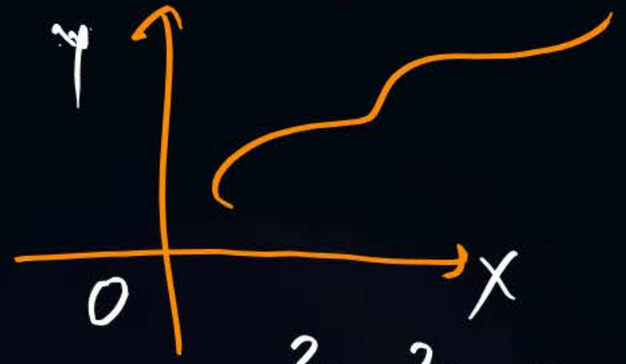
Topic

DERIVATIVES & their Types

(Part 2)

Note:

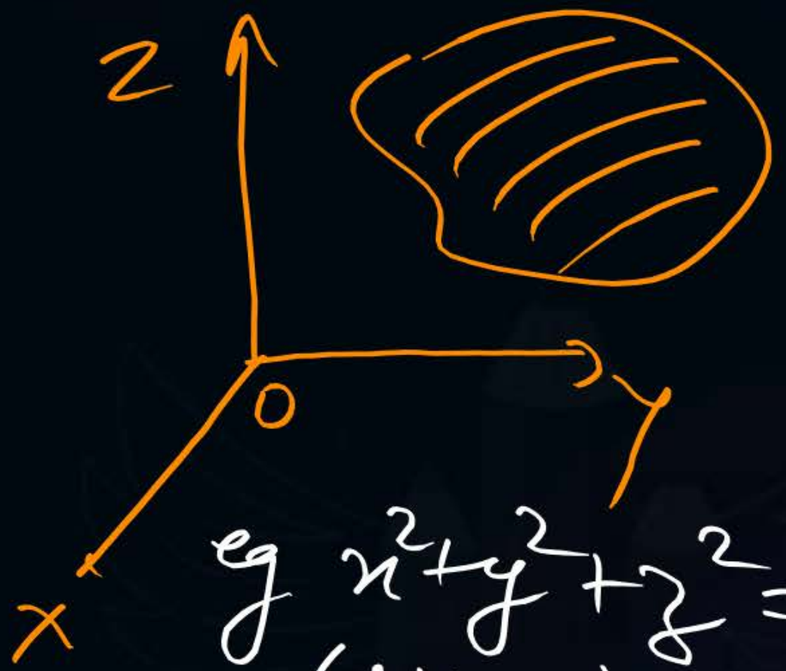
Curve: $y = f(x)$



$x^2 + y^2 = 9$

(circle)

Surface: $z = f(x, y)$



eg $x^2 + y^2 + z^2 = 9$

(Sphere)

Types of Derivatives →

Ordinary Derivative → exist in case of curve i.e $y = f(x)$

Partial Derivative → " " " surface i.e $z = f(x, y)$

Total Derivative → if $z \to (x, y) \to t$ (alone)

Derivative by using chain Rule → $z \to (x, y) \to (r, s)$

Derivative by using Jacobian Concept → if $(u, v) \to (x, y)$

then $\frac{\partial(u,v)}{\partial(x,y)}$ is called Jacobian

Explicit func^n: If it is possible to separate Dependent and Independent Variables then func^n is called Explicit func^n for eg.

$y = f(x)$, $z = f(x, y)$

Explicit Curve Explicit Surface

eg $x^3 + 3x^3y^2 + 4xy^2 = 5$

$\Rightarrow (3x^3 + 4x)y^2 = 5 - x^3$

$y = \left(\frac{5 - x^3}{3x^3 + 4x}\right)^{1/2}$

Implicit func^n: If it is not possible to separate Dep and Ind Variables then func^n is called Implicit func^n.

eg $x^3 + y^3 + 2xy = 5$

$y = ??$

$f(x, y) = c$, $f(x, y, z) = c$

Implicit Curve Implicit Surface

Ordinary Derivatives → exist in Case of Curve $y=f(x)$ →

Power formula :→ $\boxed{\frac{d}{dx}(x^a)=ax^{a-1}}$

$\frac{d}{dx}(k)=k\frac{d}{dx}(x^0)=k\{0.x^{0-1}\}=0$

$\frac{d}{dx}(x)=\frac{d}{dx}(x^1)=1.x^{1-1}=1.x^0=1$

$\frac{d}{dx}(x^2)=2x^{2-1}=2x$

$\frac{d}{dx}(x^3)=3.x^{3-1}=3x^2$

Similarly $\frac{d}{dx}(x^4)=4x^3$, $\frac{d}{dx}(x^5)=5x^4$

$\frac{d}{dx}\left(\frac{1}{x}\right)=\frac{d}{dx}(x^{-1})=-1.x^{-1-1}=\frac{-1}{x^2}$

$\frac{d}{dx}\left(\frac{1}{x^2}\right)=\frac{d}{dx}(x^{-2})=-2x^{-2-1}=\frac{-2}{x^3}$

$\frac{d}{dx}(\sqrt{x})=\frac{d}{dx}(x^{1/2})=\frac{1}{2}x^{\frac{1}{2}-1}=\frac{1}{2\sqrt{x}}$

$\frac{d}{dx}\left(\frac{1}{\sqrt{x}}\right)=\frac{d}{dx}(x^{-1/2})=-\frac{1}{2}x^{\frac{1}{2}-1}=-\frac{1}{2}x^{-3/2}$

Similarly $\frac{d^n}{dx^n}(x^n)=n!$ i.e. $\boxed{D^n(x^n)=n!}$

$\frac{d^{n+1}}{dx^{n+1}}(x^n)=0$ i.e. $\boxed{D^{n+1}(x^n)=0}$

② Exponential formula: $\frac{d}{dx}a^x = a^x \log_e a$ for e $\frac{d}{dx}(e^x) = e^x$

Q: if $y = x^a + a^x + a^a + x^x$ then $\frac{dy}{dx} = ?$

Sol: $\frac{dy}{dx} = ax^{a-1} + a^x \log_e a + 0 + x^x(1+\log_e x)$

Q: $y = \log_e x + \log_a x + \log_x a + \log_x x + \log_a a$ then $\frac{dy}{dx} = ?$

Sol: $y = \log_e x + \frac{\log_e x}{\log_e a} + \frac{\log_e a}{\log_e x} + 1 + 1$

$\frac{dy}{dx} = \frac{1}{x} + \frac{1}{\log_e a}\left(\frac{1}{x}\right) + \log_e a\left[\frac{-1}{(\log_e x)^2}\left(\frac{1}{x}\right)\right] + 0 + 0$

③ $\frac{d}{dx}(x^x) = x^x(1+\log_e x)$

④ $\frac{d}{dx}(\log_e x) = \frac{1}{x}$

eg $\frac{d^2}{dx^2}(\log_e x) = \frac{d}{dx}\left(\frac{1}{x}\right) = \frac{-1}{x^2}$

eg $\frac{d}{dx}\left(\frac{1}{\log_e x}\right) = \frac{-1}{(\log_e x)^2}\frac{d}{dx}(\log x)$

$= \frac{-1}{(\log_e x)^2}\left(\frac{1}{x}\right)$

(5) Chain Rule : $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

eg $\frac{d}{dx}\left(\sin\sqrt{\tan x^3}\right) = ?$

$= \cos\sqrt{\tan x^3}\ \frac{d}{dx}\left(\sqrt{\tan x^3}\right)$

$= \cos\sqrt{\tan x^3} \cdot \frac{1}{2\sqrt{\tan x^3}}\ \frac{d}{dx}\tan(x^3)$

$= \cos\sqrt{\tan x^3} \cdot \frac{1}{2\sqrt{\tan x^3}} \cdot \sec^2(x^3)\ \frac{d}{dx}(x^3)$

$= \downarrow \qquad (3x^2)$

Note :

$\frac{d}{dx}(xy) = x\frac{d}{dx}(y) + y\frac{d}{dx}(x)$

$= x\frac{dy}{dx} + y(1)$

$\frac{d}{dy}(xy) = x\frac{d}{dy}(y) + y\frac{d}{dy}(x)$

$= x(1) + y\frac{dx}{dy}$

$d(xy) = x\,d(y) + y\,d(x)$

$= x\,dy + y\,dx$

eg $\frac{d}{dx}(y^3) = \frac{d}{dy}(y^3)\cdot\frac{dy}{dx} = 3y^2\frac{dy}{dx}$

Some More Standard Results :-

(6) $\frac{d}{dx}(\sin x) = \cos x$

(7) $\frac{d}{dx}(\cos x) = -\sin x$

(8) $\frac{d}{dx}(\tan x) = \sec^2 x$

(9) $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

(10) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

(11) $\frac{d}{dx}(\sec x) = \sec x \tan x$

(12) $\frac{d}{dx}(\sin hx) = \cos hx$ (13) $\frac{d}{dx}(\cos hx) = \sin hx$

(14) Product formula :-

$\frac{d}{dx}(fg) = fg' + gf'$

$\frac{d}{dx}(fgh) = f'gh + fg'h + fgh'$

(15) Quotient formula :-

$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{gf' - fg'}{(g)^2}$

or $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

(16) Log Diff :→ Q: if $y=x^x$ then $\frac{dy}{dx}=?$ = ... HW $= x^x(1+\log x)$

Q: if $x^y = e^{x-y}$ then $\frac{dy}{dx}=?$

Sol: $\log_e(x^y) = \log_e(e^{x-y})$

$y\log_e x = (x-y)\log_e e$

$y\log x = x-y$

$y(1+\log x) = x$

$y = \frac{x}{1+\log x}$

$\frac{dy}{dx} = \frac{d}{dx}\left[\frac{x}{1+\log x}\right]$

$= \frac{(1+\log x)\cdot(1) - x\left(0+\frac{1}{x}\right)}{(1+\log x)^2}$

$y' = \frac{\log x}{(1+\log x)^2}$ Ans

(17) Differentiation of Infinite Series

Q: $y=\sqrt{\sin x+\sqrt{\sin x+\sqrt{\sin x+\cdots\infty}}}$

then $\frac{dy}{dx}=?$

Sol: $y=\sqrt{\sin x+y}$

$y^2=\sin x+y$

$y^2-y=\sin x$

Diff both sides w.r.to x

$\frac{d}{dx}(y^2)-\frac{d}{dx}(y)=\frac{d}{dx}(\sin x)$

$\frac{d}{dy}(y^2)\cdot\frac{dy}{dx}-\frac{dy}{dx}=\cos x$

$2y\frac{dy}{dx}-\frac{dy}{dx}=\cos x$

$\frac{dy}{dx}=\frac{\cos x}{2y-1}$ Ans

HW

$y=x^{x^{x^{\cdots\infty}}}$ then $\frac{dy}{dx}=?=\frac{y^2}{x(1-y\log x)}$

(18) Differentiation of parametric func^n

Curve	Parametric Coordinates	Equation
(1) Circle	$(a\cos\theta, a\sin\theta)$	$x^2+y^2=a^2$
(2) Parabola	$(at^2, 2at)$	$y^2=2ax$
(3) Ellipse	$(a\cos\theta, b\sin\theta)$	$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$
(4) Hyperbola	$(a\sec\theta, b\tan\theta)$	$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

(*) $\cos 2\theta = \begin{cases} \cos^2\theta - \sin^2\theta \\ 2\cos^2\theta - 1 \\ 1 - 2\sin^2\theta \end{cases}$

$\cos^2\theta - \sin^2\theta \rightarrow \frac{1+\cos 2\theta}{2} = \cos^2\theta$

$1 - 2\sin^2\theta \rightarrow \frac{1-\cos 2\theta}{2} = \sin^2\theta$

(*) if $x=f(t)$, $y=g(t)$ then $\frac{dy}{dx}=?=\frac{dy}{dt}\cdot\frac{dt}{dx}=g'(t)\left(\frac{1}{f'(t)}\right)$

$\frac{dx}{dt}=f'(t)$, $\frac{dy}{dt}=g'(t)$

Q: if $x=at^2$, $y=2at$ then $\frac{dy}{dx}=?$

Sol: $\frac{dx}{dt}=a(2t)$, $\frac{dy}{dt}=2a$

Now $\frac{dy}{dx}=\frac{dy}{dt}\cdot\frac{dt}{dx}$

$=2a\left(\frac{1}{2at}\right)=\frac{1}{t}$

Q: if $x=a(\theta-\sin\theta)$, $y=a(1-\cos\theta)$, $\frac{dy}{dx}=?$

Sol: $\frac{dx}{d\theta}=a[1-\cos\theta]$, $\frac{dy}{d\theta}=a(0+\sin\theta)$

$\frac{dy}{dx}=\frac{dy}{d\theta}\cdot\frac{d\theta}{dx}=a\sin\theta\left(\frac{1}{a(1-\cos\theta)}\right)$

$=\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}=\cot\left(\frac{\theta}{2}\right)$

(19) Differentiation of func^n w.r. to another func^n →

if $u = f(x)$, $v = g(x)$ then $\frac{du}{dv} = ? = \frac{du/dx}{dv/dx} = \frac{f'(x)}{g'(x)}$

Q. Differentiate x^x w.r. to $x \log x$?

Sol: Let $u = x^x \Rightarrow \frac{du}{dx} = \frac{d}{dx}(x^x) = x^x(1 + \log_e x)$

& let $v = x \log x \Rightarrow \frac{dv}{dx} = \frac{d}{dx}(x \log x) = x\left(\frac{1}{x}\right) + \log x\,(1) = 1 + \log x$

Hence $\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{x^x(1+\log x)}{(1+\log x)} = x^x$ Ans

Q. Differentiate $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r. to $\tan^{-1}x$?, $-1<x<1$

Note:

$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx}\cos^{-1}x = \frac{-1}{\sqrt{1-x^2}}$

$\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

$\frac{d}{dx}(\cot^{-1}x) = \frac{-1}{1+x^2}$

$\sin^{-1}x + \cos^{-1}x = \pi/2$

$\tan^{-1}x + \cot^{-1}x = \pi/2$

Sol: Put $x = \tan\theta$

Then $u = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) = \sin^{-1}(\sin 2\theta) = 2\theta$

$v = \tan^{-1}x = \theta$

Now $\frac{d\left(\sin^{-1}\left(\frac{2x}{1+x^2}\right)\right)}{d(\tan^{-1}x)} = \frac{du}{dv} = \frac{du/d\theta}{dv/d\theta} = \frac{2}{1}$

$= 2$ Ans

PARTIAL DIFF. [for $z = f(x,y)$]

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \left[\frac{f(x+h, y) - f(x,y)}{h} \right]_{y = \text{Const.}}$$

$$\frac{\partial z}{\partial y} = \lim_{k \to 0} \left[\frac{f(x, y+k) - f(x,y)}{k} \right]_{x = \text{Const.}}$$

Note:

In 2-D

Equ^n of X axis : $y = 0$

Equ^n of line $\parallel^r$ to X axis : $y = k$

Equ^n of Y axis : $x = 0$

Equ^n of line $\parallel^r$ to Y axis : $x = h$

In 3D

Equ^n of XY plane : $z = 0$

" " YZ plane : $x = 0$

" " ZX plane : $y = 0$

Equ^n of plane $\parallel^r$ to XZ plane : $y = k$

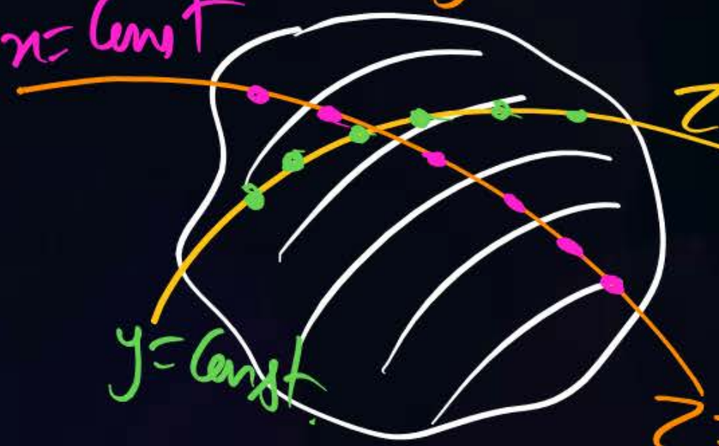
" " $\parallel^r$ to YZ plane : $x = h$

Significance of $\frac{\partial z}{\partial x}$ → $z = f(x, y)$

If we cut our surface by the plane ⊥r to x z plane (i.e form $y = \text{const}$) then we will get a curve of the type $z = f(x)$ & now

$\frac{\partial z}{\partial x}$ = slope of tangent at any Random point on this curve.

Similarly we can define $\frac{\partial z}{\partial y} = ?$



$z = f(x)$ (curve) ⇒ slope of tangent on this curve

$z = f(y)$ (curve) ⇒ slope of tangent on this curve

Note: $z = f(x,y)$ then $z_x = \frac{\partial z}{\partial x} = f_x = \frac{\partial f}{\partial x}$ all are same

(1) & $z_{xx} = f_{xx} = \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 f}{\partial x^2}$

& $z_{xy} = f_{xy} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = ?$

(2) $\boxed{\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}}$

(3) All the Results that are applicable in Case of ordinary Derivatives are also Valid in Case of Partial Derivatives keeping other Variable Constant

(4) Don't assume Dependent Variable as Constant, if we are solving Questions Based on Partial Derivatives.

Q: If $f = y^x$ then $\left(\frac{\partial^2 f}{\partial x \partial y}\right)_{\substack{x=2 \\ y=1}} = ?$

Sol: $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(y^x) = x.y^{x-1}$ $\left[\because \frac{d}{dx}(x^a) = a x^{a-1}\right]$

$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(x.y^{x-1}\right)$ $\left[\because \frac{d}{dx}(fg) = fg' + f'g\right]$

$= x.\frac{\partial}{\partial x}(y^{x-1}) + y^{x-1}\frac{\partial}{\partial x}(x)$

$\frac{\partial^2 f}{\partial x \partial y} = x\left[y^{x-1}\log_e y\right] + y^{x-1}(1)$ $\left(\because \frac{d}{dx}a^x = a^x \log_e a\right)$

$\left(\frac{\partial^2 f}{\partial x \partial y}\right)_{\substack{x=2 \\ y=1}} = 2\left[1^{2-1}\log 1\right] + 1^{2-1} = 0 + 1 = 1$ Ans

(M-II) $f = y^x$ then $\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(y^x) = y^x \log_e y$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left[y^x \cdot \log y\right]$$

$$= y^x \frac{\partial}{\partial y}(\log y) + \log y \frac{\partial}{\partial y}(y^x)$$

$$\frac{\partial^2 f}{\partial y \partial x} = y^x\left(\frac{1}{y}\right) + \log y\left(x y^{x-1}\right)$$

$$\Rightarrow \left(\frac{\partial^2 f}{\partial y \partial x}\right)_{\substack{x=2 \\ y=1}} = \frac{1^2}{1} + 0 = 1 \text{ Ans}$$

Q: if $r^2 = x^2+y^2+z^2$ then evaluate $\frac{\partial r}{\partial x}, \frac{\partial r}{\partial y}, \frac{\partial r}{\partial z}$

Sol: $r = f(x,y,z)$

Diff both sides partially w.r.to x

$$\frac{\partial}{\partial x}(r^2) = \frac{\partial}{\partial x}(x^2+y^2+z^2)$$

$$\frac{\partial}{\partial r}(r^2)\frac{\partial r}{\partial x} = 2x+0+0$$

$$(2r)\frac{\partial r}{\partial x} = 2x$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

Similarly $r^2 = x^2+y^2+z^2$

$$\frac{\partial}{\partial y}(r^2) = \frac{\partial}{\partial y}(x^2+y^2+z^2)$$

$$2r\frac{\partial r}{\partial y} = 0+2y+0$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

Similarly $\frac{\partial r}{\partial z} = \frac{z}{r}$

Q: GATE If $u=(x^2+y^2+z^2)^{-\frac{1}{2}}$ then $u_{xx}+u_{yy}+u_{zz}=?$

Sol: $u_x=\frac{\partial u}{\partial x}=\frac{\partial}{\partial x}(x^2+y^2+z^2)^{-\frac{1}{2}}=-\frac{1}{2}(x^2+y^2+z^2)^{-\frac{1}{2}-1}\frac{\partial}{\partial x}(x^2+y^2+z^2)$

$=-\frac{1}{2}(x^2+y^2+z^2)^{-3/2}(2x+0+0)$

$u_x=-x\cdot(x^2+y^2+z^2)^{-3/2}$

$u_{xx}=\frac{\partial^2 u}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)=\frac{\partial}{\partial x}\left[-x(x^2+y^2+z^2)^{-3/2}\right]$

$=-\left[x\left\{\frac{-3}{2}(x^2+y^2+z^2)^{-5/2}\cdot(2x)\right\}+(x^2+y^2+z^2)^{-3/2}(1)\right]$

(a) $2(x^2+y^2+z^2)^{-3/2}$

(b) 4.5

(c) 0

(d) $6(x^2+y^2+z^2)^{-5/2}$

$$\text{i.e. } u_{xx} = +3x^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$\text{Similarly, } u_{yy} = 3y^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$\& \quad u_{zz} = 3z^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$u_{xx}+u_{yy}+u_{zz} = 3(x^2+y^2+z^2)(x^2+y^2+z^2)^{-5/2} - 3(x^2+y^2+z^2)^{-3/2}$$

$$= 3(x^2+y^2+z^2)^{-3/2} - 3(x^2+y^2+z^2)^{-3/2}$$

$$= 0$$

(M-II) $u=(x^2+y^2+z^2)^{-1/2}$ then $u_{xx}+u_{yy}+u_{zz}=?$

(To Develop neurons)

Let $r^2=x^2+y^2+z^2$ then $\frac{\partial r}{\partial x}=\frac{x}{r}$, $\frac{\partial r}{\partial y}=\frac{y}{r}$, $\frac{\partial r}{\partial z}=\frac{z}{r}$

$u=(r^2)^{-1/2}=\frac{1}{r}$ $\Rightarrow u_x=\frac{\partial u}{\partial x}=\frac{\partial}{\partial x}\left(\frac{1}{r}\right)=\frac{\partial}{\partial r}\left(\frac{1}{r}\right)\cdot\frac{\partial r}{\partial x}$

$=-\frac{1}{r^2}\cdot\left(\frac{x}{r}\right)=\frac{-x}{r^3}$

Now, $u_{xx}=\frac{\partial^2 u}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{-x}{r^3}\right)^{*}=\frac{-1}{r^3}$ $=-\left[\frac{r^3\frac{\partial}{\partial x}(x)-x\frac{\partial}{\partial x}(r^3)}{(r^3)^2}\right]$

$=-\left[\frac{r^3(1)-x\cdot\left(3r^2\cdot\frac{\partial r}{\partial x}\right)}{r^6}\right]=\left[\frac{r^3-x\left(3r^2\cdot\frac{x}{r}\right)}{r^6}\right]$

$$\text{i.e. } U_{xx} = -\left[\frac{r^3 - 3x^2 r}{r^6}\right] = \frac{-1}{r^3} + \frac{3x^2}{r^5}$$

$$\text{Similarly, } U_{yy} = \frac{-1}{r^3} + \frac{3y^2}{r^5}$$

$$\& \quad U_{zz} = \frac{-1}{r^3} + \frac{3z^2}{r^5}$$

$$\text{i.e. } U_{xx} + U_{yy} + U_{zz} = \frac{-3}{r^3} + \frac{3(x^2+y^2+z^2)}{r^5}$$

$$= \frac{-3}{r^3} + \frac{3r^2}{r^5} = 0$$

Note: $\frac{\partial}{\partial x}(r^3) = \frac{\partial}{\partial r}(r^3)\frac{\partial r}{\partial x}$

$= 3r^2\left(\frac{x}{r}\right) = 3r.x$

Shortcut: if $u = f(r)$ & $r = f(x, y, z)$ then

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r} f'(r)$$

eg $u = (x^2 + y^2 + z^2)^{-1/2}$ then $u_{xx} + u_{yy} + u_{zz} = ?$

Sol: let $u = (r^2)^{-1/2} = \frac{1}{r} = f(r)$

$f'(r) = -1/r^2$

$f''(r) = \frac{2}{r^3}$

where $r^2 = x^2 + y^2 + z^2$.

Hence by above formula, $u_{xx} + u_{yy} + u_{zz} = \left(\frac{2}{r^3}\right) + \frac{2}{r}\left(\frac{-1}{r^2}\right) = 0$

HW if $r = x^2 + y - z$ & $y^3 + z^3 + yz - xy = 1$

where x & y are Independent Variables then at $(2, -1, 1)$

evaluate $\frac{\partial r}{\partial x} = ?$ (Ans = 4.5)

Sol: HINT: use point No. 4 of Notes.

THANK - YOU

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